Project : Linux Server Administration & Automation

1. Project Objective

Objective:
To manage and automate Linux servers by performing user and permission management, package installation, disk partitioning, backup automation, job scheduling, log management, system monitoring, and troubleshooting to ensure security, stability, and optimal performance.

2. Tools & Technologies Used

Tool / Technology	Purpose
Linux OS (Ubuntu / CentOS)	Server administration
Shell Commands	System operations
Cron	Job scheduling
tar, gzip	Backup & compression
fdisk, mkfs	Disk partitioning

3. User & Group Management

Purpose:
To control user access and improve system security.

Commands & Explanation

`useradd user1`

Creates a new user named `user1`. `passwd`

`user1`

Sets a password for `user1`. `groupadd`

`admin`

Creates a group named `admin`. `usermod`

`-aG admin user1`

Adds `user1` to the `admin` group.

`id user1`

```
[linux12@vbox ~]$ su - root
Password:
[root@vbox ~]#
[root@vbox ~]# useradd vijay
```

```
[root@vbox ~]# ls
anaconda-ks.cfg  demo  etc1.tar  files
[root@vbox ~]# cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
games:x:12:100:games:/usr/games:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/sbin/nologin
tss:x:59:59:Account used for TPM access:/:/usr/sbin/nologin
systemd-coredump:x:999:999:systemd Core Dumper:/:/sbin/nologin
dbus:x:81:81:System message bus:/:/sbin/nologin
polkitd:x:998:998:User for polkitd:/:/sbin/nologin
sssd:x:997:997:User for sssd:/:/sbin/nologin
avahi:x:70:70:Avahi mDNS/DNS-SD Stack:/var/run/avahi-daemon:/sbin/nologin
colord:x:996:996:User for colord:/var/lib/colord:/sbin/nologin
geoclue:x:995:995:User for geoclue:/var/lib/geoclue:/sbin/nologin
rtkit:x:172:172:RealtimeKit:/:/sbin/nologin
setroubleshoot:x:994:993:SELinux troubleshoot server:/var/lib/setroubleshoot:/usr/sbin/nologin
clevis:x:993:991:Clevis Decryption Framework unprivileged user:/var/cache/clevis:/usr/sbin/nologin
flatpak:x:992:990:Flatpak system helper:/:/usr/sbin/nologin
libstoragemgmt:x:989:989:daemon account for libstoragemgmt:/:/usr/sbin/nologin
pipewire:x:988:988:PipeWire System Daemon:/run/pipewire:/usr/sbin/nologin
gdm:x:42:42:GNOME Display Manager:/var/lib/gdm:/usr/sbin/nologin
gnome-initial-setup:x:987:987::/run/gnome-initial-setup/:/sbin/nologin
chrony:x:986:986:chrony system user:/var/lib/chrony:/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/usr/sbin/nologin
dnsmasq:x:985:985:Dnsmasq DHCP and DNS server:/var/lib/dnsmasq:/usr/sbin/nologin
tcpdump:x:72:72::/:/sbin/nologin
linux12:x:1000:1000:linux12:/home/linux12:/bin/bash
vijay:x:1001:1001::/home/vijay:/bin/bash
```

```
vijay:x:1001:1001::/home/vijay:/bin/bash
[root@vbox ~]# passwd vijay
Changing password for user vijay.
New password:
BAD PASSWORD: The password contains the user name in some form
Retype new password:
passwd: all authentication tokens updated successfully.
[root@vbox ~]# groupadd devops
[root@vbox ~]# cat /etc/gropus
cat: /etc/gropus: No such file or directory
[root@vbox ~]# cat /etc/group
```

Displays user ID, group ID, and group membership.

4. File & Permission Management

Purpose:
To restrict unauthorized access to files and directories.

Commands & Explanation

`mkdir /project_data`

Creates a directory for project data.

`chown user1:admin /project_data`

Changes ownership to user `user1` and group `admin`. `chmod`

`770 /project_data`

Provides full access to owner and group, no access to others.

`ls -ld /project_data`

Verifies directory permissions.

```
[root@vbox ~]# mkdir /project_data
[root@vbox ~]# ls
anaconda-ks.cfg  demo  etc1.tar  files
[root@vbox ~]# ll
total 26456
-rw-------. 1 root root     1094 Feb 17 14:23 anaconda-ks.cfg
drwxr-xr-x. 2 root root        6 Feb 17 16:39 demo
-rw-r--r--. 1 root root 27084800 Feb 17 16:42 etc1.tar
drwxr-xr-x. 2 root root        6 Feb 17 15:31 files
[root@vbox ~]# cd project_data
-bash: cd: project_data: No such file or directory
[root@vbox ~]# cd
[root@vbox ~]# ls
anaconda-ks.cfg  demo  etc1.tar  files
[root@vbox ~]# ll
total 26456
-rw-------. 1 root root     1094 Feb 17 14:23 anaconda-ks.cfg
drwxr-xr-x. 2 root root        6 Feb 17 16:39 demo
-rw-r--r--. 1 root root 27084800 Feb 17 16:42 etc1.tar
drwxr-xr-x. 2 root root        6 Feb 17 15:31 files
[root@vbox ~]# cd project_data
-bash: cd: project_data: No such file or directory
[root@vbox ~]#
[root@vbox ~]# mkdir /project_data
mkdir: cannot create directory '/project_data': File exists
[root@vbox ~]# chown vijay:devops /project_data
[root@vbox ~]# ls
anaconda-ks.cfg  demo  etc1.tar  files
[root@vbox ~]# ls -ld
dr-xr-x---. 6 root root 4096 Feb 19 10:29 .
[root@vbox ~]# ls -ld /project_data
drwxr-xr-x. 2 vijay devops 6 Feb 19 10:32 /project_data
[root@vbox ~]# chmod 770 /project_data
[root@vbox ~]# ls -ld /project_data
drwxrwx---. 2 vijay devops 6 Feb 19 10:32 /project_data
[root@vbox ~]# apt update
```

5. Package Management

Purpose:
To install and manage required software packages. **Ubuntu**

/ Debian

```
apt update apt install nginx
zip unzip -y
```

CentOS / RHEL

```
yum install nginx zip unzip -y
```

These commands install web server and compression tools.

```
bash: apt: command not found...
[root@vbox ~]# yum install nginx zip unzip cronie -y
CentOS Stream 9 - BaseOS
3.3 kB/s | 4.5 kB     00:01
CentOS Stream 9 - AppStream
6.0 kB/s | 4.6 kB     00:00
CentOS Stream 9 - Extras packages
 15 kB/s | 4.9 kB     00:00
Package zip-3.0-35.el9.x86_64 is already installed.
Package unzip-6.0-59.el9.x86_64 is already installed.
Package cronie-1.5.7-14.el9.x86_64 is already installed.
Dependencies resolved.
================================================================================================================================
 Package                                    Architecture                  Version                                   Reposito
ry                        Size
================================================================================================================================
Installing:
 nginx                                      x86_64                        2:1.20.1-26.el9                           appstrea
m                         36 k
Upgrading:
 cronie                                     x86_64                        1.5.7-16.el9                              baseos
                         119 k
 cronie-anacron                             x86_64                        1.5.7-16.el9                              baseos
                          31 k
Installing dependencies:
 centos-logos-httpd                         noarch                        90.9-1.el9                                appstrea
m                        1.5 M
 nginx-core                                 x86_64                        2:1.20.1-26.el9                           appstrea
m                        571 k
 nginx-filesystem                           noarch                        2:1.20.1-26.el9                           appstrea
m                        9.3 k

Transaction Summary
================================================================================================================================
Install  4 Packages
```

6. Disk Partitioning & Permanent Mount

Purpose:
To manage additional storage effectively.

Commands & Explanation

```
lsblk
```

Lists available disks and partitions.

```
fdisk /dev/sdb
```

Creates a new partition on disk `/dev/sdb`. `mkfs.ext4`

`/dev/sdb1`

Formats the partition using the ext4 filesystem.

`mkdir /data`

Creates a mount directory. `mount`

`/dev/sdb1 /data`

Mounts the partition temporarily. `blkid`

Displays UUID of the partition.

```
[root@vbox ~]# blkid
/dev/mapper/cs_vbox-root: UUID="f05dcb58-7088-46b8-af21-0d679e7a6b30" TYPE="xfs"
/dev/sda2: UUID="96QuyA-DoYq-ehME-g7k9-9uOY-wVdl-oLhigL" TYPE="LVM2_member" PARTUUID="8632b187-02"
/dev/mapper/cs_vbox-swap: UUID="2ff65ed2-bb06-4c76-8655-6e64e681b175" TYPE="swap"
/dev/sdb2: PARTUUID="9402a082-02"
/dev/sdb3: PARTUUID="9402a082-03"
/dev/sdb1: PARTUUID="9402a082-01"
/dev/sda1: UUID="96e5bb12-8542-4e6f-aa60-8aa2f1e4a01e" TYPE="xfs" PARTUUID="8632b187-01"
[root@vbox ~]# lsblk
NAME             MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                8:0    0   20G  0 disk
├─sda1             8:1    0    1G  0 part /boot
└─sda2             8:2    0   19G  0 part
  ├─cs_vbox-root 253:0    0   17G  0 lvm  /
  └─cs_vbox-swap 253:1    0    2G  0 lvm  [SWAP]
sdb                8:16   0   10G  0 disk
├─sdb1             8:17   0    1G  0 part
├─sdb2             8:18   0    2G  0 part
├─sdb3             8:19   0    2G  0 part
└─sdb4             8:20   0    1K  0 part
sr0               11:0    1 1024M  0 rom
[root@vbox ~]# fdisk /dev/sdb

Welcome to fdisk (util-linux 2.37.4).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.


Command (m for help): n
All primary partitions are in use.
Adding logical partition 5
First sector (10489856-12584959, default 10489856):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (10489856-12584959, default 12584959): +1G
```

```
bash: mkfs.exf4: command not found...
[root@vbox ~]# mkfs.ext4 /dev/sdb1
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 262144 4k blocks and 65536 inodes
Filesystem UUID: 6f2144a4-b98d-4c15-8b79-94bbeb039209
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

[root@vbox ~]# mkfs.
mkfs.cramfs  mkfs.ext2    mkfs.ext3    mkfs.ext4    mkfs.fat     mkfs.minix   mkfs.msdos   mkfs.vfat    mkfs.xfs
[root@vbox ~]# mkfs.ext4 /dev/s
sda       sda2      sdb1      sdb3      sdb5      sg1       shm/      snd/      stderr    stdout
sda1      sdb       sdb2      sdb4      sg0       sg2       snapshot  sr0       stdin
[root@vbox ~]# mkfs.ext4 /dev/s
sda       sda2      sdb1      sdb3      sdb5      sg1       shm/      snd/      stderr    stdout
sda1      sdb       sdb2      sdb4      sg0       sg2       snapshot  sr0       stdin
[root@vbox ~]# mkfs.ext4 /dev/s
sda       sda2      sdb1      sdb3      sdb5      sg1       shm/      snd/      stderr    stdout
sda1      sdb       sdb2      sdb4      sg0       sg2       snapshot  sr0       stdin
[root@vbox ~]# mkfs.ext4 /dev/s
sda       sda2      sdb1      sdb3      sdb5      sg1       shm/      snd/      stderr    stdout
sda1      sdb       sdb2      sdb4      sg0       sg2       snapshot  sr0       stdin
[root@vbox ~]# mkfs.ext4 /dev/s
sda       sda2      sdb1      sdb3      sdb5      sg1       shm/      snd/      stderr    stdout
sda1      sdb       sdb2      sdb4      sg0       sg2       snapshot  sr0       stdin
[root@vbox ~]# mkfs.ext4 /dev/s
sda       sda2      sdb1      sdb3      sdb5      sg1       shm/      snd/      stderr    stdout
sda1      sdb       sdb2      sdb4      sg0       sg2       snapshot  sr0       stdin
[root@vbox ~]# mkfs.ext4 /dev/s
sda       sda2      sdb1      sdb3      sdb5      sg1       shm/      snd/      stderr    stdout
sda1      sdb       sdb2      sdb4      sg0       sg2       snapshot  sr0       stdin
[root@vbox ~]# mkfs.ext4 /dev/s
sda       sda2      sdb1      sdb3      sdb5      sg1       shm/      snd/      stderr    stdout
sda1      sdb       sdb2      sdb4      sg0       sg2       snapshot  sr0       stdin
[root@vbox ~]# mkfs.ext4 /dev/s
sda       sda2      sdb1      sdb3      sdb5      sg1       shm/      snd/      stderr    stdout
sda1      sdb       sdb2      sdb4      sg0       sg2       snapshot  sr0       stdin
[root@vbox ~]# mkfs.ext4 /dev/s
sda       sda2      sdb1      sdb3      sdb5      sg1       shm/      snd/      stderr    stdout
sda1      sdb       sdb2      sdb4      sg0       sg2       snapshot  sr0       stdin
[root@vbox ~]# mkfs.ext4 /dev/s
sda       sda2      sdb1      sdb3      sdb5      sg1       shm/      snd/      stderr    stdout
sda1      sdb       sdb2      sdb4      sg0       sg2       snapshot  sr0       stdin
[root@vbox ~]# mkfs.ext4 /dev/s
sda       sda2      sdb1      sdb3      sdb5      sg1       shm/      snd/      stderr    stdout
```

Permanent Mount Configuration

`vi /etc/fstab` Add

entry:

```
UUID=xxxxxx   /data   ext4   defaults   0   0 mount
-a
```

Validates the permanent mount.

```
        created on Thu Feb 19 10:51:39 2026
Proceed anyway? (y,N) mkfs

[root@vbox ~]# mkdir /data
[root@vbox ~]# mount /dev/sdb1 /data
[root@vbox ~]# blkid
/dev/mapper/cs_vbox-root: UUID="f05dcb58-7088-46b8-af21-0d679e7a6b30" TYPE="xfs"
/dev/sda2: UUID="96QuyA-DoYq-ehME-g7k9-9uOY-wVdl-oLhigL" TYPE="LVM2_member" PARTUUID="8632b187-02"
/dev/mapper/cs_vbox-swap: UUID="2ff65ed2-bb06-4c76-8655-6e64e681b175" TYPE="swap"
/dev/sda1: UUID="96e5bb12-8542-4e6f-aa60-8aa2f1e4a01e" TYPE="xfs" PARTUUID="8632b187-01"
/dev/sdb1: UUID="6f2144a4-b98d-4c15-8b79-94bbeb039209" TYPE="ext4" PARTUUID="9402a082-01"
/dev/sdb2: PARTUUID="9402a082-02"
/dev/sdb5: PARTUUID="9402a082-05"
/dev/sdb3: PARTUUID="9402a082-03"
[root@vbox ~]# lsblk
NAME               MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                  8:0    0   20G  0 disk
├sda1                8:1    0    1G  0 part /boot
└sda2                8:2    0   19G  0 part
  ├cs_vbox-root 253:0    0   17G  0 lvm  /
  └cs_vbox-swap 253:1    0    2G  0 lvm  [SWAP]
sdb                  8:16   0   10G  0 disk
├sdb1                8:17   0    1G  0 part /data
├sdb2                8:18   0    2G  0 part
├sdb3                8:19   0    2G  0 part
├sdb4                8:20   0    1K  0 part
└sdb5                8:21   0 1023M  0 part
sr0                 11:0    1 1024M  0 rom
[root@vbox ~]# blkid
/dev/mapper/cs_vbox-root: UUID="f05dcb58-7088-46b8-af21-0d679e7a6b30" TYPE="xfs"
/dev/sda2: UUID="96QuyA-DoYq-ehME-g7k9-9uOY-wVdl-oLhigL" TYPE="LVM2_member" PARTUUID="8632b187-02"
/dev/mapper/cs_vbox-swap: UUID="2ff65ed2-bb06-4c76-8655-6e64e681b175" TYPE="swap"
/dev/sda1: UUID="96e5bb12-8542-4e6f-aa60-8aa2f1e4a01e" TYPE="xfs" PARTUUID="8632b187-01"
/dev/sdb1: UUID="6f2144a4-b98d-4c15-8b79-94bbeb039209" TYPE="ext4" PARTUUID="9402a082-01"
/dev/sdb2: PARTUUID="9402a082-02"
/dev/sdb5: PARTUUID="9402a082-05"
/dev/sdb3: PARTUUID="9402a082-03"
[root@vbox ~]# lsblk
NAME               MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                  8:0    0   20G  0 disk
├sda1                8:1    0    1G  0 part /boot
└sda2                8:2    0   19G  0 part
  ├cs_vbox-root 253:0    0   17G  0 lvm  /
  └cs_vbox-swap 253:1    0    2G  0 lvm  [SWAP]
sdb                  8:16   0   10G  0 disk
├sdb1                8:17   0    1G  0 part /data
├sdb2                8:18   0    2G  0 part
```

7. Backup Automation

Purpose:
To ensure data safety using automated backups.

Commands & Explanation

`mkdir /backup`

Creates backup storage directory. `tar -czvf /backup/data_$(date +%F).tar.gz /data` Creates a compressed backup of `/data`. `crontab -e`

Opens crontab editor.

`0 2 * * * tar -czvf /backup/data_$(date +\%F).tar.gz /data` Schedules daily backup at 2:00 AM.

```
[root@vbox ~]# lsblk
NAME              MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                 8:0    0   20G  0 disk
├─sda1              8:1    0    1G  0 part /boot
└─sda2              8:2    0   19G  0 part
  ├─cs_vbox-root 253:0    0   17G  0 lvm  /
  └─cs_vbox-swap 253:1    0    2G  0 lvm  [SWAP]
sdb                 8:16   0   10G  0 disk
├─sdb1              8:17   0    1G  0 part /data
├─sdb2              8:18   0    2G  0 part
├─sdb3              8:19   0    2G  0 part
├─sdb4              8:20   0    1K  0 part
└─sdb5              8:21   0 1023M  0 part
sr0                11:0    1 1024M  0 rom
[root@vbox ~]# mkdir /backup
[root@vbox ~]# tar -czvf /backup/data_$(date +%F).tar.gz /data
tar: Removing leading `/' from member names
/data/
/data/lost+found/
[root@vbox ~]# corntab-e
bash: corntab-e: command not found...
[root@vbox ~]# crontab -e
no crontab for root - using an empty one
crontab: installing new crontab
[root@vbox ~]# crontab -l
0 2 * * * tar -czvf /backup/data_$(data +\%F).tar.gz /data0 2 * * * tar -czvf /backup/data_$(data +\%F).tar.gz /data
[root@vbox ~]# crontab -e
crontab: installing new crontab
[root@vbox ~]# cat /tmp/cron_test.log
cat: /tmp/cron_test.log: No such file or directory
[root@vbox ~]# cat /tmp/data.log
cat: /tmp/data.log: No such file or directory
[root@vbox ~]# cat /sdb1/data.log
cat: /sdb1/data.log: No such file or directory
[root@vbox ~]# cat /tmp/cron_test.log
cron is working
```

8. Log Management

Purpose:
To reduce disk usage and manage logs efficiently.

```
tar  -czvf  /backup/logs.tar.gz  /var/log
```

Compresses system logs. `find /backup -type f`

```
-mtime +7 -delete
```

Deletes backup files older than 7 days.

9. Job Schedulin

Job Scheduling means **running commands or scripts automatically** at a **specific time or interval** without manual execution.

2 Types of Job Scheduling in Linux

Type	Tool
One-time job	`at`
Repeating job	`cron`
System-wide periodic jobs	`cron.*`
Startup job	`@reboot` (cron)

3 One-Time Job Scheduling (at)

Step 1: Check at service

```
systemctl status atd
```

Start if stopped:

`systemctl start atd` **Step**

2: Schedule a job

```
at 5:30 PM
```

Then type command: `echo "Hello Linux"`

```
>> /tmp/test.txt
```

Press **Ctrl + D** to save.

Step 3: View scheduled jobs

```
atq
```

Step 4: Remove a job `atrm`

```
<job_id>
```

4 Repeating Job Scheduling (cron)

Step 1: Open crontab `crontab`

```
-e
```

Step 2: Cron format

```
* * * * * command
| | | | |
| | | | └─ Day of week (0-7)
| | | └── Month (1-12)
| | └─── Day of month (1-31)
| └──── Hour (0-23) └──────
Minute (0-59)
```

Step 3: Example cron jobs

Run every minute

```
* * * * * echo "Cron test" >> /tmp/cron.txt
```

Run daily at 2 AM

```
0 2 * * * /root/backup.sh
```

Run every Sunday

```
0 10 * * 0 /root/weekly.sh
```

Step 4: View cron jobs

```
crontab -l
```

Step 5: Remove cron jobs

```
crontab -
```

5 System-Wide Cron Jobs

File/Directory	Purpose
`/etc/crontab`	System cron
`/etc/cron.hourly/`	Hourly jobs
`/etc/cron.daily/`	Daily jobs
`/etc/cron.weekly/`	Weekly jobs
`/etc/cron.monthly/`	Monthly jobs

6 Run Job at System Boot

```
@reboot /root/startup.sh
```

(Add inside `crontab -e`)

7 Allow / Deny Scheduling

File	Purpose
`/etc/cron.allow`	Allowed users
`/etc/cron.deny`	Denied users

`/etc/at.allow` Allowed users

`/etc/at.deny` Denied users

8 Difference: cron vs at

cron	at
Repeated jobs	One-time job
Time-based	Single execution
Runs continuously	Executes once

```
 -V          print version and exit
 -x <mask>   enable debugging

Default operation is replace, per 1003.2
[root@vbox ~]# crontab -e
crontab: installing new crontab
[root@vbox ~]# atq
3       Thu Feb 19 14:16:00 2026 a root
[root@vbox ~]# crontab -e
crontab: installing new crontab
[root@vbox ~]# atq
3       Thu Feb 19 14:16:00 2026 a root
[root@vbox ~]# atq
[root@vbox ~]# crontab -l
0 16 19 3 5 /root/backup.sh
25 14 19 3 5 /root/backup.sh
0 2 * * 0 /root/backup.sh
0 15 19 3 5 /root/backup.sh
 0 2 * * * tar -czvf /backup/data_$(data +\%F).tar.gz /data0 2 * * * tar -czvf /backup/dat
a_$(data +\%F).tar.gz /data
 * * * * * echo "cron is working " >> /tmp/cron_test.log

[root@vbox ~]#
```

10. System Monitoring

Purpose:

To monitor server performance and resource usage. `top`

Displays real-time CPU and process usage. `free`

`-h`

Shows memory usage. `df`

`-h`

Displays disk usage.

`systemctl status nginx`

```
1     PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   6155 linux12    20   0  775140  52824  40788 S   0.3   3.0   0:02.56 gnome-terminal-
      1 root       20   0  183324  17340  11100 S   0.0   1.0   0:01.46 systemd
      2 root       20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root       20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_
2    PID    PPID     TIME+  %CPU  %MEM  PR  NI S    VIRT    RES    UID COMMAND
   6849       2  0:00.00   0.0   0.0  20   0 I       0      0      0 kworker/u4:3-event+
   6729       2  0:00.13   0.0   0.0  20   0 I       0      0      0 kworker/0:0-events
   6707       2  0:00.11   0.0   0.0  20   0 I       0      0      0 kworker/u4:2-event+
   6589       2  0:00.04   0.0   0.0  20   0 I       0      0      0 kworker/u4:0-event+
3    PID  %MEM    VIRT    RES   CODE   DATA    SHR nMaj nDRT  %CPU COMMAND
   5431  17.4 3571976 303384      8 286184 128652  139    0   0.0 gnome-shell
   5763   5.3  972320  92308    468 118004  46760  251    0   0.0 gnome-software
   6155   3.0  775140  52824    184  67912  40788   41    0   0.3 gnome-terminal-
    775   2.5  352556  43208      4  32668  19636  218    0   0.0 firewalld
4    PID   PPID   UID USER     RUSER    TTY          TIME+  %CPU  %MEM S COMMAND
    734      1   172 rtkit    rtkit    ?          0:00.05   0.0   0.2 S rtkit-daemon
      1      0     0 root     root     ?          0:01.46   0.0   1.0 S systemd
      2      0     0 root     root     ?          0:00.00   0.0   0.0 S kthreadd
      3      2     0 root     root     ?          0:00.00   0.0   0.0 S pool_workqueue_
      4      2     0 root     root     ?          0:00.00   0.0   0.0 I kworker/R-rcu_+
[root@vbox ~]# free -h
              total        used        free      shared  buff/cache   available
Mem:          1.7Gi       883Mi       382Mi        20Mi       615Mi       824Mi
Swap:         2.0Gi       0.0Ki       2.0Gi
[root@vbox ~]#
```

```
   5763   5.3  972320  92308    468 118004  46760  251    0   0.0 gnome-software
   6155   3.0  775140  52824    184  67912  40788   41    0   0.3 gnome-terminal-
    775   2.5  352556  43208      4  32668  19636  218    0   0.0 firewalld
4    PID   PPID   UID USER     RUSER    TTY          TIME+  %CPU  %MEM S COMMAND
    734      1   172 rtkit    rtkit    ?          0:00.05   0.0   0.2 S rtkit-daemon
      1      0     0 root     root     ?          0:01.46   0.0   1.0 S systemd
      2      0     0 root     root     ?          0:00.00   0.0   0.0 S kthreadd
      3      2     0 root     root     ?          0:00.00   0.0   0.0 S pool_workqueu
      4      2     0 root     root     ?          0:00.00   0.0   0.0 I kworker/R-rcu
[root@vbox ~]# free -h
              total        used        free      shared  buff/cache   available
Mem:          1.7Gi       883Mi       382Mi        20Mi       615Mi       824Mi
Swap:         2.0Gi       0.0Ki       2.0Gi
[root@vbox ~]# df -h
Filesystem                Size  Used Avail Use% Mounted on
devtmpfs                  821M     0  821M   0% /dev
tmpfs                     854M     0  854M   0% /dev/shm
tmpfs                     342M  6.3M  336M   2% /run
/dev/mapper/cs_vbox-root   17G  6.3G   11G  38% /
/dev/sda1                 960M  439M  522M  46% /boot
tmpfs                     171M   96K  171M   1% /run/user/1000
[root@vbox ~]# df -sh /data
df: invalid option -- 's'
Try 'df --help' for more information.
[root@vbox ~]# du -sh /data
0       /data
[root@vbox ~]#
```

```
tmpfs                       342M  6.3M  336M   2% /run
/dev/mapper/cs_vbox-root     17G  6.3G   11G  38% /
/dev/sda1                   960M  439M  522M  46% /boot
tmpfs                       171M   96K  171M   1% /run/user/1000
[root@vbox ~]# df -sh /data
df: invalid option -- 's'
Try 'df --help' for more information.
[root@vbox ~]# du -sh /data
0       /data
[root@vbox ~]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 100
0
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group defaul
t qlen 1000
    link/ether 08:00:27:dc:a1:6a brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 84852sec preferred_lft 84852sec
    inet6 fd17:625c:f037:2:a00:27ff:fedc:a16a/64 scope global dynamic noprefixroute
       valid_lft 86206sec preferred_lft 14206sec
    inet6 fe80::a00:27ff:fedc:a16a/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[root@vbox ~]#
```

11. Troubleshooting

Purpose:
To identify and resolve system issues. `journalctl`

`-xe`

Displays detailed system error logs.

`du -ah / | sort -rh | head -10`

Identifies large files consuming disk space.

`systemctl restart nginx`

Restarts a failed service.

12. Project Outcome

- Implemented secure user and permission management
- Automated backups and log maintenance
- Improved system stability and performance
- Reduced manual administrative effort

13. Conclusion

This project provided hands-on experience in Linux server administration and automation, demonstrating essential skills required for Linux Administrator and DevOps Engineer roles.

Ravi Babasaheb Shete